Ministry of Science and Higher Education of the Russian Federation
NATIONAL RESEARCH UNIVERSITY ITMO

R E P O R T

on the laboratory work
"Investigation of the Characteristics of a DC Electrical Energy Source"

in the discipline "Electronics"

Variant 032

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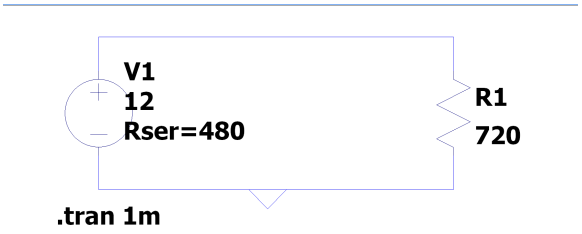
1. Objective of the work

Investigation of operating modes and experimental determination of the parameters of the equivalent circuit of a DC electrical energy source.

2. Initial data (Variant 032)

- **Source parameters:** $E = 12.000$ V, $r = 480.000$ Ω.
- **Load parameters** Rn_k **[Ω]** $(k = 0 \dots 10)$**:**
  $Rn_0 = \infty$ (R_1), $Rn_1 = 4320$ (R_2), $Rn_2 = 1920$ (R_3), $Rn_3 = 1120$ (R_4), $Rn_4 = 720$ (R_5), $Rn_5 = 480$ $(R_6 = r)$, $Rn_6 = 320$ (R_7), $Rn_7 = 206$ (R_8), $Rn_8 = 120$ (R_9), $Rn_9 = 53$ (R_{10}), $Rn_{10} = 0$ (R_{11}).

3. Experimental circuit



a) Simulation circuit in LTspice

Figure 1.1 – Experimental circuit

4. Table of experimental and calculated data

Table 1.1

Parameters of the equivalent circuit									
$r_p = 480.000$ [Ω]			$E_p = 12.000$ [V]				$J_p = 25.000$ [mA]		
k	Rn_k [Ω]	Un_k [V]	In_k [mA]	Pn_k [mW]	r_k [Ω]	In_{pk} [mA]	Un_{pk} [V]	Pn_{pk} [mW]	η_k
0	∞	12.000	0.000	0.000	—	0.000	12.000	0.000	1.000
1	4320	10.800	2.500	27.000	480.000	2.500	10.800	27.000	0.900
2	1920	9.600	5.000	48.000	480.000	5.000	9.600	48.000	0.800
3	1120	8.400	7.500	63.000	480.000	7.500	8.400	63.000	0.700
4	720	7.200	10.000	72.000	480.000	10.000	7.200	72.000	0.600
5	480	6.000	12.500	75.000	480.000	12.500	6.000	75.000	0.500
6	320	4.800	15.000	72.000	480.144	15.000	4.800	72.000	0.400
7	206	3.603	17.493	63.028	479.856	17.493	3.603	63.028	0.300
8	120	2.400	20.000	48.000	480.111	20.000	2.400	48.000	0.200
9	53	1.193	22.514	26.865	479.887	22.514	1.193	26.865	0.099
10	0	0.000	25.000	0.000	—	25.000	0.000	0.000	0.000

5. Example of measurements and calculation (experiment $k = 4$)

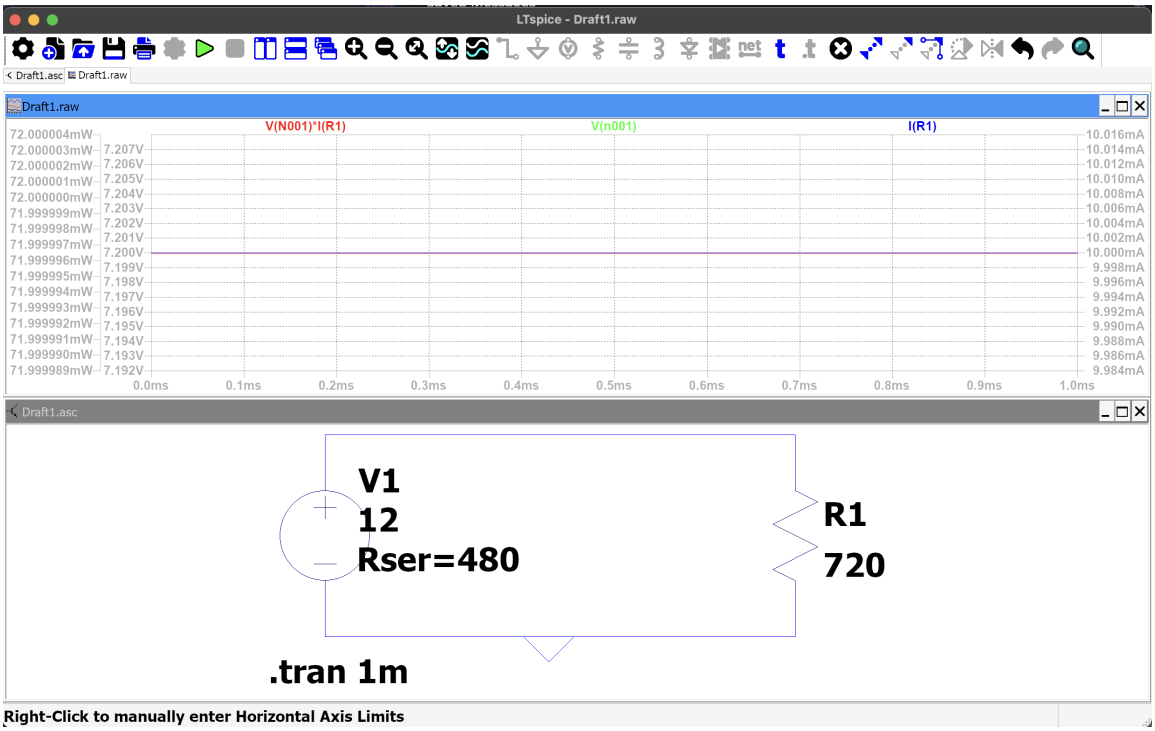


Figure 1.2 – Measurement results in LTspice for experiment $k = 4$ ($Rn_4 = 720\ \Omega$)

Initial data for calculation ($k = 4$):
$Rn_4 = 720.000\ \Omega$, $Un_4 = 7.200$ V, $Un_5 = 6.000$ V, $In_4 = 10.000$ mA $= 0.010$ A, $In_5 = 12.500$ mA $= 0.0125$ A, $E_p = 12.000$ V, $r_p = 480.000\ \Omega$.

Calculation of the equivalent circuit parameters:

$$E_p = Un_0 = 12.000\ [\text{V}]$$

$$r_p = \sqrt{\frac{\sum_{k=1}^{9} r_k^2}{9}} = \sqrt{\frac{5 \cdot 480.000^2 + 480.144^2 + 479.856^2 + 480.111^2 + 479.887^2}{9}} = 480.000\ [\Omega]$$

$$J_p = \frac{E_p}{r_p} = \frac{12.000}{480.000} = 0.025\ [\text{A}] = 25.000\ [\text{mA}]$$

Calculation relations (for $k = 4$):

1. Internal resistance of the source at section $k = 4$:

$$r_4 = \frac{Un_4 - Un_5}{In_5 - In_4} = \frac{7.200 - 6.000}{0.0125 - 0.0100} = \frac{1.200}{0.0025} = 480.000\ [\Omega]$$

2. Calculated current in the load:

$$In_{p4} = \frac{E_p}{Rn_4 + r_p} = \frac{12.000}{720.000 + 480.000} = 0.010\ [\text{A}] = 10.000\ [\text{mA}]$$

3. Calculated voltage across the load:

$$Un_{p4} = E_p - r_p \cdot In_{p4} = 12.000 - 480.000 \cdot 0.010 = 7.200\ [\text{V}]$$

4. Calculated power in the load:

$$Pn_{p4} = Un_{p4} \cdot In_{p4} = 7.200 \cdot 0.010 = 0.072 \text{ [W]} = 72.000 \text{ [mW]}$$

5. Efficiency (COP):

$$\eta_4 = 1 - \frac{r_p}{r_p + Rn_4} = 1 - \frac{480.000}{480.000 + 720.000} = 0.600$$

6. Graphs of source characteristics

6.1. External characteristic of the source $Un(In)$

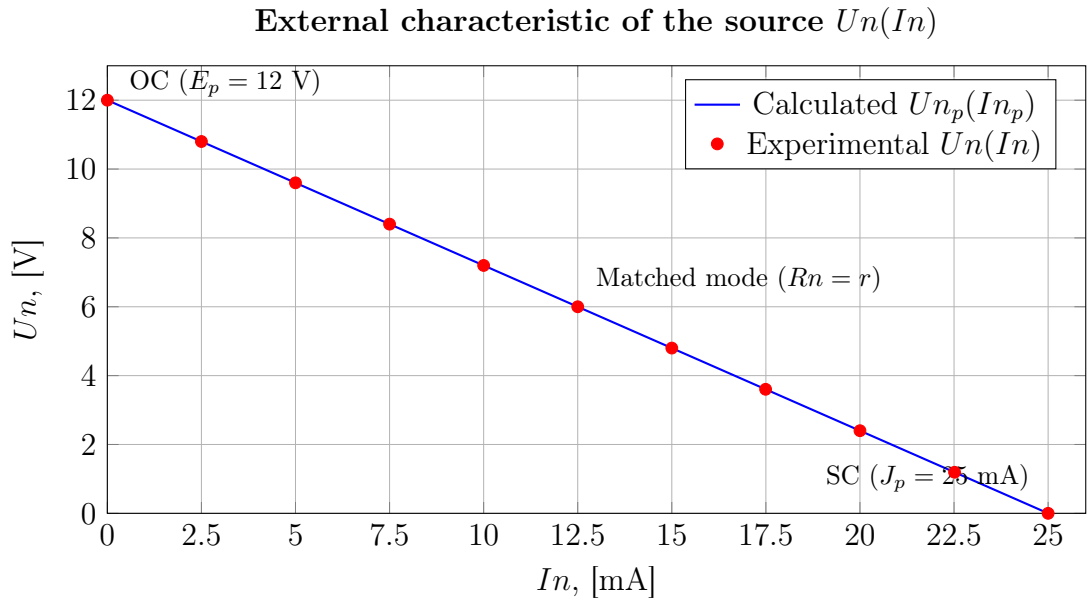


Figure 1.3 – External characteristic of the electrical energy source

6.2. Operating characteristics of the source $Pn(In)$ and $\eta(In)$

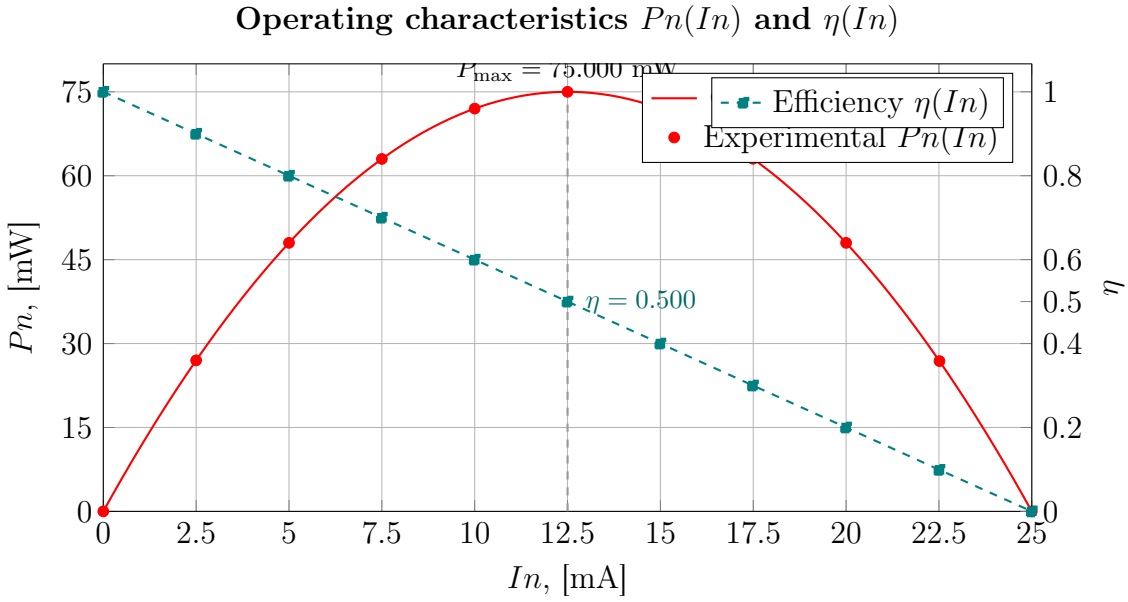


Figure 1.4 – Dependence of load power $Pn(In)$ and efficiency $\eta(In)$ on load current

7. Conclusions

In the course of this laboratory work, the operating modes of a DC electrical energy source were investigated in the LTspice circuit simulation software (Variant 032: $E = 12.000$ V, $r = 480.000$ Ω):

1. Based on the experimental results, the parameters of the equivalent circuit of the source were determined with an accuracy of thousandths: the calculated EMF of the source was $E_p = 12.000$ V, the root-mean-square internal resistance $r_p = 480.000\ \Omega$, the short-circuit current $J_p = 25.000$ mA. The calculated values fully coincide with the experimental simulation data.

2. The external characteristic of the source $Un(In)$ is linear and monotonically decreases from the open-circuit voltage $Un_0 = E_p = 12.000$ V (at $In_0 = 0.000$ mA, $Rn_0 = \infty$) to $Un_{10} = 0.000$ V in the short-circuit mode (at $In_{10} = J_p = 25.000$ mA, $Rn_{10} = 0$).

3. The dependence of the useful power $Pn(In)$ has the form of a parabola. In the extreme modes – open-circuit ($In = 0$) and short-circuit ($Un = 0$) – the useful power in the load is 0.000 mW.

4. The maximum power in the load $P_{\max} = 75.000$ mW is delivered in the matched operating mode (experiment $k = 5$) when the load resistance equals the internal resistance of the source ($Rn_5 = r_p = 480.000\ \Omega$). In this case, the load current is $In_5 = 0.5 \cdot J_p = 12.500$ mA, the terminal voltage is $Un_5 = 0.5 \cdot E_p = 6.000$ V, and the source efficiency is $\eta_5 = 0.500$ (50%). As the open-circuit mode is approached, the efficiency tends to 1.000, but the useful power tends to zero.